Ammonia emissions from mineral fertilizers – Results on mitigation potential from a multiyear and multisite experiment across Germany

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Materials and Methods

0.1. Trial design

Investigations in the joint project NH_3-Min were based on coordinated multi-plot field trials across six different German regions representing typical wheat production areas, with each region featuring up to two field sites (Figure 1). Eight different fertilizers, receiving the same site-specific yearly amount of N, as well as one unfertilized control were tested. The different treatments were allocated to a randomized complete block design with four repetitions. The unfertilized control (N0) was used to correct for background emissions within the trial. The size of the square experimental plots was 9x9 m with a 9 m buffer area between each plot fertilized with non NH_3-emitting $CaNO_3^-$ fertilizer.

All sites featured winter wheat (*Triticum aestivum* L. var. RGT Reform) and timing of seeding as well as crop protection measures were decided based on best management practices and executed by the respective research stations.

0.2. Site descriptions

Experimental sites with different environmental conditions for fertilizer performance were spread out across Germany (Figure 1). Soil samples for site characterization were taken before fertilizer application in spring. Samples were sieved, air dried and analyzed all together in the same central laboratory. The main site characteristics are given in Table 1.

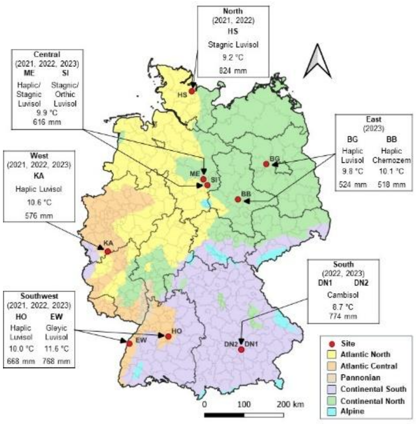


Figure 1: Geographical locations of NH_3-Min trial with respective study years, soil type and mean annual temperature and rainfall. Soil types according to world reference base for soil resources (FAO, 2014). Mean annual temperature and rainfall are based on the years 1991-2020 and were retrieved from the closest weather station of the German meteorological service (DWD, 2024). The map also highlights grouped environmental zones across Germany modified based on Metzger et al. (2005).

Table 1: Main site characteristics. Values rounded to one single digit. EW2023, DN2023 and SI2023: trial on same respective field but plots shifted to avoid overlap with 2021 trial.

Region	Site	Year	Latitude	Longitude	Texture			C_{org}	N_t	pH	pH buffer capacity	CEC	Urease activity
					Sand	Silt	Clay			$CaCl_2$			
			degr. N	degr. E	%	%	%	%	%		mmol kg^{-1}	$cmol_c\ kg^{-1}$	mg N kg^{-1} h^{-1}
North	HS	2021	54.313407	9.993680	59.4	27.6	13.0	1.1	0.11	6.9	38.7	10.6	23.5
	HS	2022	54.211040	9.983022	52.5	32.0	15.5	1.2	0.11	6.8	39.3	12.0	22.6
East	BB	2023	51.832900	11.693100	15.5	65.2	19.3	1.8	0.16	7.1	32.7	17.9	28.0
	BG	2023	52.619600	12.784900	71.6	19.9	8.5	0.8	0.09	6.4	21.9	6.9	11.0
South	DN1	2022+2023	48.407058	11.694203	13.8	64.7	21.5	1.2	0.13	6.8	23.9	13.8	23.3
	DN2	2022	48.407058	11.694203	15.9	57.0	27.1	1.3	0.14	6.5	24.2	17.8	23.9
Southwest	EW	2021+2023	48.545250	7.851324	38.6	41.3	20.1	1.0	0.11	6.8	30.5	13.9	16.1
	EW	2022	48.518299	7.869955	33.7	50.0	16.3	1.0	0.08	6.0	29.5	7.4	13.3
	HO	2021	48.710884	9.193204	8.1	68.0	23.9	1.8	0.18	6.6	39.7	16.8	37.7
	HO	2022	48.716286	9.188530	7.9	69.3	22.8	1.4	0.14	6.8	34.6	13.9	30.4
	HO	2023	48.713643	9.195351	3.2	72.0	24.8	1.1	0.13	7.1	36.2	15.6	30.4
West	KA	2021	50.616390	6.996154	6.4	75.5	18.1	1.1	0.11	6.4	31.9	11.3	21.3
	KA	2022	50.620352	6.988096	6.3	77.0	16.7	1.0	0.10	6.9	29.1	10.7	19.3
	KA	2023	50.617860	6.988583	7.4	76.0	16.6	0.9	0.10	6.7	29.5	10.7	15.1
Central	ME	2021	52.370582	10.547239	74.6	19.9	5.5	1.5	0.11	5.5	26.7	5.4	15.8
	ME	2022	52.387282	10.562512	68.6	24.2	7.2	1.4	0.11	6.6	23.8	7.0	20.5
	ME	2023	52.366386	10.491218	79.1	12.0	8.9	1.2	0.12	6.6	20.5	8.3	21.6
	SI	2021+2023	52.190478	10.686515	17.6	69.4	13.0	1.6	0.16	6.3	45.2	13.5	23.7
	SI	2022	52.202587	10.635506	48.6	37.3	14.1	1.6	0.14	6.4	33.5	13.2	24.1

0.3. Fertilizers

The fertilizers used in this study were urea (U), urea with the UI 2-NPT (U+UI), and doubled inhibited urea with 2-NPT and the NI MPA (U+UI+NI), calcium ammoniaum nitrate (CAN), U+AS (UAS), urea ammonium nitrate (UAN), UAN + UI NBPT/NPPT (® Limus) and injected UAN+AS solution (Cultan method CU) All fertilizers are commercially available. According to the producer, the concentration of the active ingredient 2-NPT in treatments U+UI and U+UI+NI accounted for 0.075% of the N-content of the fertilizers. For the U+UI+NI fertilizer the corresponding MPA concentration was 0.1% of the fertilizer N-content. The fertilizers were obtained every spring from agricultural distributors, to avoid inhibitor degradation during storage over several years.

The annual amount of fertilizer applied was indicated by the site specific maximum given by the German fertilizer ordinance (DüV, 2020). Spring soil mineral N (0 - 0.9 m depth) as well as potential site-specific mineralization during the vegetation period was subtracted from the N requirement for a five-year average wheat yield at the experimental sites (DüV, 2020). The fertilizer amounts differed across sites and years because of diverging yield potentials, pre-crops and soil organic carbon concentrations. Fertilizer application was split into three applications for U and U+UI. The first N application was applied in early spring, during tillering of the wheat plants (ca. BBCH 23). During stem elongation (ca. BBCH 32) the second split application was applied. The third application was applied between ear emergence and flowering (ca. BBCH 49). The ratio between the three split applications was determined by the respective research stations. For U+UI+NI the same annual amount of fertilizer was applied as for the other treatments but it was split evenly only over the first two applications, expect for region north in 2021 with three applications. In region south, all fertilizers were applied only in two split applications. For each NH_3 measurement campaign, fertilizers were applied at the same day using a tractor mounted pneumatic fertilizer spreader or a box fertilizer spreader.

0.4. NH_3 measurement

0.4.1. Measurement method

The NH_3-Min multi plot field trials used the method of calibrated passive sampling (CPS) for emission calculation (Pacholski, 2016), by using Inverse Dispersion Modelling (IDM) in combination with Alpha Samplers for calculation of absolute ammonia fluxes recently validated in a study by (Götze et al., 2025). As passive samplers (Vandré and Kaupenjohann, 1998) only give qualitative treatment differences, they need to be scaled by a quantitative method (Pacholski, 2016). Employing IDM (Loubet et al., 2018; Nikolajsen et al., 2020) in combination with ALPHA samplers (Tang et al., 2001) absolute NH_3 emissions were estimated from atmospheric NH_3 concentrations, wind speed and wind direction at 2 m height. Passive samplers were used in every plot, ALPHA samplers only in the highest emitting U treatment. Additionally, in three out of the four unfertilized control plots ALPHA samplers were used to capture the background emissions. The described method of IDM with ALPHA samplers was recently described by Götze et al. (2025). This is the first paper using this novel approach for comparison of NH_3 emissions with different fertilizer treatments on large coordinated field trials across Germany.

0.4.2. Management of NH_3 samplers in the field and laboratory

Triplicates of ALPHA samplers and one passive sampler were fastened on steel rods 0.25 m above the canopy in the center of the respective plot. Sampling intervals were between approximately 24 h when emissions occurred and up to 72 h at the end of a measurement campaign with very low emissions and atmospheric NH_3 concentrations. Measurement campaigns lasted from one to three weeks, with both types of samplers changed simultaneously. The passive samplers consisted of 0.25 l square bottom PE bottles with two 0.02 m diameter holes on each side. The holes were covered with a fine mesh, to allow for easy air exchange and to exclude insects and debris. For NH_3 measurement the samplers were filled with 0.02 l of 0.05 *M* sulfuric acid. After exposition, the volume of the remaining solution in the samplers was sampled, quantified and the samples were then frozen until analysis. ALPHA samplers sample NH_3 diffusing through a PTFE membrane by citric acid coated-filter paper. After exposition, the filter paper was removed from the ALPHA sampler body and frozen until analysis. For determination of NH_4^+-N uptake, the ALPHA filter papers were extracted with 0.005 l of deionized water. For NH_4^+-N measurement of the extracts and sulfuric acid solutions of the passive samplers NH_3 selective electrodes or colorimetric analyzers were employed. A list of NH_4^+-N measurement devices at the individual sites is given in **Fehler! Verweisquelle konnte nicht gefunden werden.**. An interlaboratory test confirmed the accuracy and comparability of the individual laboratories and measurement devices.

0.4.3. Emissions estimation from ALPHA samplers by inverse dispersion modelling

Atmospheric NH_3 concentration of ALPHA samplers was calculated based on the effectively sampled air volume and the amount of absorbed N (Tang et al., 2001). Inverse dispersion modelling was done using a backward Lagrangian stochastic (bLS) model (Flesch et al., 2004). The "WindTrax" software (Thunder Beach Scientific, Nova Scotia, Canada), mapped the experimental plot as an emission source and the ALPHA sampler as a concentration sensor. Using wind speed, friction velocity and atmospheric stability, the bLS model estimates half-hourly fluxes. The actual ALPHA emission is the result of dividing the measured atmospheric NH_3 concentration by the simulated concentration to emission ratio.

The process of IDM modelling and the derivation of input parameters for the WindTrax software is described in detail in Götze et al. (2025). When compared to integrated horizontal flux measurements as a micrometeorological reference method (Kemmann et al., 2025) Götze et al. (2025) recently confirmed that this approach, including a correction for low wind speed and high temperature conditions, resulted in accurate measurements. This comparison of NH_3 measurement methods was part of the NH_3-Min project to validate small scale replicate plot NH_3 loss measurements against the integrated horizontal flux method as a reference and was integrated in the field trials described here.

0.4.4. Derivation of transfer coefficient

In calibrated passive sampling a correction of semi-quantitative passive sampler measurements is necessary by quantitative emission data. This scaling was done in past studies by a simple transfer coefficient (TC) calculation. The TC is the quotient which is obtained by dividing absolute NH_3 emissions by passive sampler concentrations. However, background emissions have a substantial influence on TC calculation. As the NH_3-Min field trials represented a large diversity of such conditions, TC calculation was not always successful. Comparatively small passive sampler concentrations or high ALPHA sampler emissions led to biased TCs, thus overestimating NH_3 emissions. The measurement campaigns in slow emitting synthetic fertilizers tend to last longer compared to organic fertilizers, where the concept of calibrated passive sampling was developed. To improve standard TC calculation, multiple statistical models were tested on a subset of six site-year combinations. The models were fitted separately for each measurement campaign. The root mean square error between the model predicted emissions and the input measured emissions was used to assess the goodness of fit. The campaign specific linear mixed model (Equation 1) gave the lowest root means square errors across campaigns.

Equation 1: Linear mixed model for transfer coefficient estimation.

$$y_{ijklm} = \mu_{klm} + \beta_{klm}x_{ijklm} + \beta_{jklm}x_{ijklm} + \rho_{iklm} + e_{ijklm}$$

Where y_{ijklm} is the absolute emission (kg ha^{-1}) in the i^{th} plot on the j^{th} day of the k^{th} measurement campaign at the l^{th} site in the m^{th} year. μ_{klm} is the general intercept for the k^{th} measurement campaign at the l^{th} site in the m^{th} year. β_{klm} is the fixed slope for the k^{th} measurement campaign at the l^{th} site in the m^{th} year. x_{ijklm} is the measured passive sampler concentrations (mg N l^{-1}) in the i^{th}

plot on the j^{th} day of the k^{th} measurement campaign at the l^{th} site in the m^{th} year. β_{jklm} is a random day specific slope for the j^{th} day of the k^{th} measurement campaign at the l^{th} site in the m^{th} year. ρ_{iklm} is a random plot effect for the i^{th} plot in the k^{th} measurement campaign at the l^{th} site in the m^{th} year. e_{ijklm} is the random error of y_{ijklm}.

The general intercept μ and the fixed slope β explain the main passive sampler influence on absolute emissions. The random day specific slope β_i gives an estimate on the influence of day specific meteorological conditions on the relationship between passive samplers and NH_3 emissions based on ALPHA samplers connected with IDM. The random plot effect estimation was not feasible for plots without ALPHA samplers and was set to zero in these plots. Comparable to a standard TC, the proposed linear mixed model was fitted individually for each NH_3 measurement campaign, using the "lmer" function from the R-package "lme4" (Bates et al., 2015). The function "predict" from the R-package "stats" was used to calculate absolute emissions from passive sampler concentrations (R Core Team, 2023). The standard error of absolute emissions was estimated by 10,000-fold bootstrapping using the function "bootMer" from the R-package "lme4" (Bates et al., 2015).

This novel approach on TC calculation based on a linear mixed model has the following advantages: i) The placement of the plots within the multi-plot trial and thus the associated background emissions are considered in the calculation. ii) A factor for the day-specific correlation between ALPHA IDM emissions and passive sampler NH_3 uptake is included in the model. This is especially important for long measurement campaigns under changing meteorological conditions, as expected for emission trials with inhibited synthetic fertilizers. iii) The prediction error can be considered as the standard error of the emissions for further calculations, rendering the need of error propagation calculation during ALPHA IDM unnecessary.

Compared to standard TC calculation, a mixed model-based TC greatly improved the prediction accuracy of absolute NH_3 emissions within the calibrated passive sampling approach in our study and can be suggested for future studies. We opted not to include further variables such as site and weather conditions into the TC model. As weather variables influence NH_3 emissions for both types of samplers, we focused on the influence of site and weather conditions on the absolute emissions, using statistical analysis (See 0.6.).

0.5. Weather conditions

Air temperature, precipitation and wind speed were determined by weather stations at every field site. Wind speed was additionally measured at the same height as the alpha samples. Temperature and wind speed were averaged per campaign. Standardized rain (mm d^{-1}) was calculated from cumulative rainfall for each campaign divided by its duration in days, providing a measure for the rainfall intensity. In case of missing site-specific weather data, gaps were filled from nearby stations of the German meteorological service (DWD).

0.6. Statistical analysis

0.6.1. Outline

Daily emissions were cumulated for each plot as well as for measured background emissions in the control plots. The control treatment was submitted to the model similar to the other treatments. This approach was chosen because control subtraction from treatments before modeling would result in correlation of background subtracted emission, voiding a fundamental premise of statistical evaluation. Due to the unbalanced nature of the data between sites, treatments and years as well as the repeated measuring of emissions on the same plots during one experimental year, a linear mixed effects model was used for analysis of the data. Normality of the data was checked visually using QQ-plots of the residuals. As the emission data was not normally distributed, the data was log transformed before statistical analysis. After fitting of the mixed model and covariable selection, back-transformed estimates were obtained to subtract the background emissions from treatment emissions. Input related emissions were calculated based on background subtracted emissions and fertilized N-amount per treatment. Significant differences between treatments on the back-transformed scale were indicated by the log scale p-values of the pairwise differences.

0.6.2. Linear mixed effects model for emissions

To analyze the influence of fertilizer treatment and covariables on NH_3-emissions, the base line mixed model given in Equation 2 was fitted.

Equation 2: Linear mixed effects model for analysis of NH_3-emissions.

$$\eta_{ijklm} = \mu + \tau_i + \alpha_j + (\tau\alpha)_{ij} + Y_k + L_l + B_{m(lk)} + (\tau Y)_{ik} + (\tau L)_{il} + (YL)_{kl} + (\tau YL)_{ikl} + e_{ijklm}$$

where η_{ijklm} is the log-transformed NH_3-emission cumulated per campaign in the *j*-th campaign from the *i*-th treatment applied in the *m*-th block of the at the *l*-th location and in the *k*-th year. Here, μ is the overall mean, τ_i is the main effect of the *i*-th treatment, α_j is the main effect of the *j*-th campaign, $(\tau\alpha)_{ij}$ is the interaction effect of the *i*-th treatment with *j*-th campaign, Y_k is the main effect of the *k*-th year, L_l is the main effect of the *l*-th location, $B_{m(lk)}$ is the main effect of the *m*-th block effect nested within the *l*-th location in the *k*-th year, $(\tau Y)_{ik}$ and $(\tau L)_{il}$ is the interaction effects of the *i*-th treatment with *k*-th year and *l*-th location, respectively, $(YL)_{kl}$ is the interaction effect of the *l*-th location and *k*-th year, $(\tau YL)_{ikl}$ is the interaction effect of the *i*-th treatment with *l*-th location and *k*-th year, and e_{ijklm} is the random plot error associated with η_{ijklm}. Here, treatment effect τ_i, campaign effect α_j and treatment by campaign interaction $(\tau\alpha)_{ij}$ were fitted as fixed effects and remainder of the effects in model (Equation 2) were fitted as random effects with $Y_k \sim N(0, \sigma_Y^2)$, $L_l \sim N(0, \sigma_L^2)$, $B_{m(nlk)} \sim N(0, \sigma_B^2)$, $(\tau Y)_{ik} \sim N(0, \sigma_{\tau Y}^2)$, $(\tau L)_{il} \sim N(0, \sigma_{\tau L}^2)$, $(\tau YL)_{ikl} \sim N(0, \sigma_{\tau YL}^2)$. As multiple NH_3 measurements were collected from the same experimental plots in one year, the residual covariance structure of the model was set to account for autocorrelation of experimental plots across campaigns. A heterogenous autoregressive model of order 1 [ARH(1)] was fitted which allowed the determination of a campaign-specific residual variance.

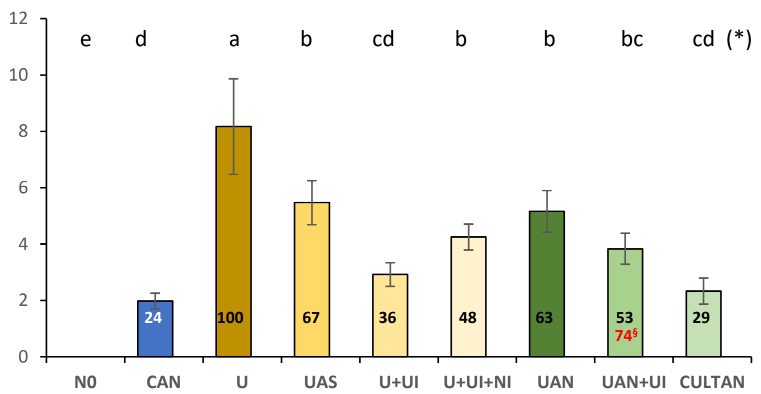
To incorporate the environmental covariates $(x_1, \ldots, x_p)$, the model (Equation 2) was extended, and treatment specific regressions $(\beta_1, \ldots, \beta_p)$ on each covariate and their interaction were fitted. The available covariates were soil properties (soil pH, clay content, total organic carbon, CEC, pH buffer capacity, urease activity, soil water content, soil NH_4-N concentration) and weather data (average temperature, average windspeed, standardized rain). The soil covariates were location-specific, the weather covariates were campaign specific. All covariates were standardized to mean 0 and standard deviation 1, ensuring stable estimates for explanatory variables. Assumptions of the model were checked using the diagnostic plots. Stepwise process for model selection was used, and the model with the lowest AIC from the full likelihood was selected. However, the final selected model was fitted using the residual maximum likelihood (REML). Since the treatment campaign interaction $(\tau\alpha)_{ij}$ was no longer significant after covariable selection, it was removed from the final model.

Results

Model averaged emissions and emission reductions across years and study sites for each fertilizer application timing

Fertilizer	Fertilizer split application*	n	Relative emission	Emission reduction
	I start of vegetation, II tillering, III ear emergence		(% N applied)	compared to surface applied granular urea (%)
CAN	I	88	3.3±1.2	72.7±7.5
CAN	II	84	1.6±0.4	72.7±7.5
CAN	III	80	2±0.8	72.7±7.5
CU	I	36	4±1.5	67.2±12.2
CU	II	35	1.9±0.5	67.2±12.2
CU	III	36	2.4±1	67.2±12.2
U	I	88	12.2±4.4	
U	II	84	5.8±1.3	
U	III	80	7.2±3	
U+UI	I	87	4.7±1.7	61±8
U+UI	II	83	2.3±0.5	61±8
U+UI	III	79	2.8±1.2	61±8
U+UI+NI	I	51	5.8±2.2	52±12
U+UI+NI	II	47	2.8±0.7	52±12
U+UI+NI	III	8	3.4±1.5	52±12
UAN	I	88	7.7±2.8	36.5±9.1
UAN	II	84	3.7±0.8	36.5±9.1
UAN	III	79	4.6±1.9	36.5±9.1
UAN+UI	I	82	5.7±2.1	53±8.4
UAN+UI	II	77	2.7±0.6	53±8.4
UAN+UI	III	75	3.4±1.4	53±8.4
UAS	I	88	8.6±3.1	29.4±9.4
UAS	II	84	4.1±0.9	29.4±9.4
UAS	III	80	5.1±2.1	29.4±9.4

cumulative NH_3-emission (% fertilizer N)



§ emission relative to UAN as reference

References

DüV, 2020. Verordnung über die Anwendung von Düngemitteln, Bodenhilfsstoffen, Kultursubstraten und Pflanzenhilfsmitteln nach den Grundsätzen der guten fachlichen Praxis beim Düngen vom 10. August 2021 (BGBl. I S. 3436).

Götze, H., Brokötter, J., Frößl, J., Kelsch, A., Kukowski, S., Pacholski, A.S., 2025. Assessment of different methods to determine NH3 emissions from small field plots after fertilization. Environments 12, 255. https://doi.org/10.3390/environments12080255.

Kemmann, B., Brokötter, J., Götze, H., Kelsch, A., Frößl, J., Riesch, S., Heinemann, P., Kukowski, S., Pacholski, A.S., Flessa, H., 2025. Ammonia emissions from urea fertilization – Multi-annual micrometeorological measurements across Germany. Agric. Ecosyst. Environ. 381, 109416. https://doi.org/10.1016/j.agee.2024.109416.

Loubet, B., Carozzi, M., Voylokov, P., Cohan, J.-P., Trochard, R., Génermont, S., 2018. Evaluation of a new inference method for estimating ammonia volatilisation from multiple agronomic plots. Biogeosciences 15, 3439–3460. https://doi.org/10.5194/bg-15-3439-2018.

Nikolajsen, M.T., Pacholski, A.S., Sommer, S.G., 2020. Urea ammonium nitrate solution treated with inhibitor technology: Effects on ammonia emission reduction, wheat yield, and inorganic N in soil. Agronomy 10, 161. https://doi.org/10.3390/agronomy10020161.

Pacholski, A.S., 2016. Calibrated Passive Sampling--Multi-plot field measurements of NH3 emissions with a combination of dynamic tube method and passive samplers. JoVE - Journal of visualized experiments. https://doi.org/10.3791/53273.

Piepho, H.-P., 2004. An algorithm for a letter-based representation of all-pairwise comparisons. J. Comput. Graph. Stat. 13, 456–466. https://doi.org/10.1198/1061860043515.

Piepho, H.-P., Malik, W.A., Bischoff, R., El-Hasan, A., Scheer, C., Sedlmeier, J.E., Gerhards, R., Petschenka, G., Voegele, R.T., 2024. Efficacy assessment in crop protection: a tutorial on the use of Abbott's formula. J. Plant. Dis. Prot. 131, 2139–2160. https://doi.org/10.1007/s41348-024-00968-0.

R Core Team, 2023. R: A Language and Environment for Statistical Computing. Vienna, Austria. https://www.R-project.org/.

Tang, Y.S., Cape, J.N., Sutton, M.A., 2001. Development and types of passive samplers for monitoring atmospheric NO2 and NH3 concentrations. Sci. World J. 1, 513–529. https://doi.org/10.1100/tsw.2001.82.

Vandré, R., Kaupenjohann, M., 1998. In situ measurement of ammonia emissions from organic fertilizers in plot experiments. Soil Sci. Soc. Am. J. 62, 467. https://doi.org/10.2136/sssaj1998.03615995006200020025x.